

KENAR NARAYAKA

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EDUCATION

University of Oxford – Christ Church

Oxford, Oxfordshire, England

MCompPhil, Computer Science & Philosophy - Predicted 1st Class

Oct. 2025 – Jun. 2029

Relevant Coursework: Probability, Linear Algebra, Logic, Discrete Maths, Design & Analysis of Algorithms, Algorithm & Data Structures, Continuous Maths, Functional Programming, Imperative Programming, Intro. Proof Systems

EXPERIENCE

Hitachi Astemo

Jun. 2024 – Aug. 2024

Backend Engineer Intern | *PHP, Fat-Free Framework (F3), Postgres, Docker*

Jakarta, Indonesia

- Replaced paper-based HR processes covering ~15,000 employee records with browser-based administrative tools, translating legacy forms, approval states, and organisational relationships into normalised Postgres schemas.
- Designed and Dockerised REST APIs and a multi-stage approval engine with server-side validation, role-based access control, immutable audit history, and fulfilment tracking; reduced end-to-end request turnaround from approximately 7 days to 5 hours, a 97% improvement
- Implemented permission-scoped CRUD dashboards and coordinate-based PDF generation connected to network printers, then extracted repeated schema mappings, form components, and printable-template logic into reusable tooling for subsequent internal workflows.

CapitOx

Jan. 2025 – Apr. 2026

Quant Academy – Oxford-Man Institute | *Python, NumPy, SciPy, Matplotlib*

- Selected for a competitive CapitOx programme run with the Oxford-Man Institute, covering mean-variance portfolio construction, factor models, signal combination, and the design and evaluation of multi-signal systematic trading strategies.
- Built a Monte Carlo pricing pipeline for a European put under a 2-component generalised-normal mixture model, implementing numerically stable log-density evaluation and separating the target distribution, proposal mechanism, and estimator logic for controlled comparison.
- Implemented 3 estimators with proposal tuning, convergence diagnostics, and reproducible seeded experiments; evaluated them across 100+ trials and achieved 5x lower variance and 60x lower runtime at matched error relative to the baseline random-walk Metropolis–Hastings implementation.

PROJECTS

3D Fractal Simulator | *JavaScript, three.js, GLSL*

A-Level NEA Project

- Built a real-time browser-based renderer for Mandelbulb, Juliabulb, and Mandelbox geometries, combining signed-distance estimation with custom GLSL ray marching, configurable lighting and colour pipelines, and free-camera controls exposed through an interactive three.js interface.
- Implemented distance-adaptive step sizing that used estimated scene geometry to reduce unnecessary samples and increase frame rate by ~40%, then added cone marching for a further ~20% improvement over the original fixed-step pipeline.
- Profiled rendering performance across camera distances, iteration counts, and fractal parameters to balance throughput against close-range geometric detail; awarded 69/70, the highest OCR Computer Science NEA score in the school's history

Basketball Jumpshot Analyser | *Tauri v2, Rust, ONNX Runtime, FFmpeg*

- Built a fully offline desktop app that analyses basketball jump shots from a single video, designed as a Rust analysis core behind a Tauri v2 + Svelte shell so all processing is done locally; ran top-down RTMPose pose estimation through ONNX Runtime with a hand-written SimCC keypoint decoder.
- Decoded footage through FFmpeg keyed on true frame timestamps to stay accurate on variable-frame-rate and 120–240 fps slow-motion clips, then segmented continuous multi-shot recordings into per-rep phases and extracted 28 biomechanical parameters across six kinematic families.
- Fit a gravity-constrained release parabola in closed-form least squares and aligned each rep to a proportion-scaled ideal model via phase-anchored dynamic time warping; ranked coaching cues by shot-to-shot consistency behind a typed availability model that reports metrics as unavailable rather than fabricating single-view 3-D quantities.

TECHNICAL SKILLS

Programming Languages: Python, C++, PHP, JavaScript, SQL, GLSL, Rust

Frameworks & Libraries: NumPy, SciPy, Matplotlib, three.js, Fat-Free Framework, Tauri v2

Tools: Postgres, Git, Linux, tmux, Docker

Concepts: Backend Development, Database Design, Graphics Programming, REST APIs